

Characterization of Geometric Space via Homotopy Type Theory and Gnomonic Growth

Formal Research Draft

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Abstract

This paper analyzes the derivation of the intrinsic properties of a geometric space $\mathcal{S}$ within the framework of Homotopy Type Theory (HoTT). By employing the J-rule and modal logic, we demonstrate how the intersection of dependent types evolves toward infinite dimensions through gnomonic growth functions.

Formal Framework

- **Setup:** $u : A, \quad B : \mathcal{U}, \quad q := A \cap B$
- **Evolution:** $t_0 : \mathcal{U} \rightarrow (\mathbf{f}_{esp} \times 3 \rightarrow t_1)$
- **Limit:** $p = \infty\text{-dim} \implies (p \rightarrow G)$ where $(\mathbf{f}_{esp} > \mathbf{f}_{gno})$
- **J-Rule:** $P(u) = p(\mathcal{U}) \vdash J(P, r, u, p) : P(u) \rightarrow p(\mathcal{U})$
- **Transport:** $f(\text{tr}(A)) \rightarrow \dim_1(u) \rightarrow \cdots \rightarrow \dim_n(u) \rightarrow p$
- **Sum Axiom:** $\sum_{(u:\mathcal{U})} \mathbf{f}_{gno}(q) \implies \Box \mathbf{f}_{gno}$
- **Modal A (Necessity):** $A = ((\dim_n \wedge t_n) \rightarrow \Box \mathbf{f}_{gno})$
- **Modal B (Possibility):** $B = ((\dim_n \wedge t_n) \rightarrow q) \wedge (B \rightarrow \Diamond(\mathbf{f}_{gno} \rightarrow A(u)))$

1 Introduction

Geometric space $\mathcal{S}$ is not a static container but a higher-order type. The structure defined in the formal framework above shows that every point u is an inhabitant of a dynamic intersection q .

2 Analysis of Gnomonic Growth

The function $\mathbf{f}_{gno}$ acts as a transport constraint. In the proposed derivation, gnomonic growth is the fundamental element allowing the transition from finite dimension 0 to infinite dimension p . This is not a stochastic event but is governed by the necessity operator $\Box$.

3 Derivation of Infinite Dimension

By applying the **J-rule** (identity elimination), we transport the equality between types along the temporal path. If u exists at t_0 , the internal geometry of the space—defined by the intersection $A \cap B$ —forces the existence of a gnomonic witness at every subsequent stage t_n .

4 Modal Interpretation

The distinction between types A and B lies in their logical nature:

- **Type A** represents the rigid structure of the dimension (Always Necessary).
- **Type B** represents the potentiality of the space to intersect with A to generate new dimensions (Always Possible).

5 Conclusion

This paper proves that the gnomonic intersection is the foundation for the stability of an infinite-dimensional space within the HoTT context. The coherence of the system is guaranteed by the transport of u across the dimensions $0 \rightarrow n \rightarrow \infty$.